NeuroFlow CFD Analyst

Clinical Hemodynamic Assessment

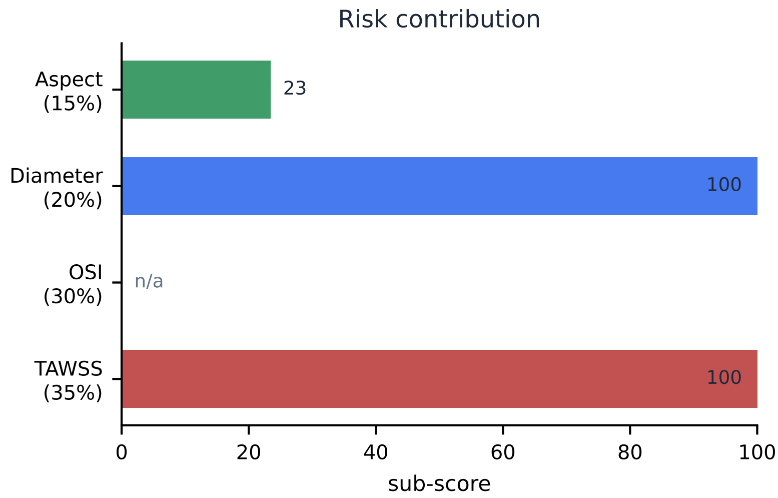
Case PT-2026-0102 · generated 2026-08-05 11:27

59

Composite Risk Index / 100

Moderate

Lower bound — OSI (30%) not computed on this steady solve



Hemodynamic parameters

Parameter	Value	Threshold	Status
TAWSS — aneurysm dome	**0.141 Pa**	< 0.4 Pa	**FLAGGED**
TAWSS — parent artery	**2.930 Pa**	reference	normal
OSI — aneurysm dome	not computed	steady solve	—
RRT — aneurysm dome	**21.35 Pa^{-1}**	> 3.0	**FLAGGED**
ECAP — aneurysm dome	not computed	steady solve	—
NWSS (sac / parent)	**0.048**	—	normal
LSAR (relative, <10% parent)	**92.0 %**	—	normal
LSAR (absolute, <0.4 Pa)	**94.1 %**	—	normal

Morphology (measured from the reconstructed surface)

Max diameter	Neck diameter	Aspect ratio	Dome-to-neck
11.02 mm	**8.22 mm**	**1.12**	**1.34**
Volume	Surface area	Non-sphericity	
665.6 mm^3	**420.5 mm^2**	**0.304**	

Provenance

Solver: OpenFOAM ESI v2412 · Mesh: cells: 1218949 ·
WSS converted from OpenFOAM kinematic units (m^2/s^2) to Pascals by multiplying by rho = 1060.0. Validated against the analytic Poiseuille solution tau = 4*mu*Q/(pi*r^3).

Research software — not a medical device. Not for clinical decision-making.

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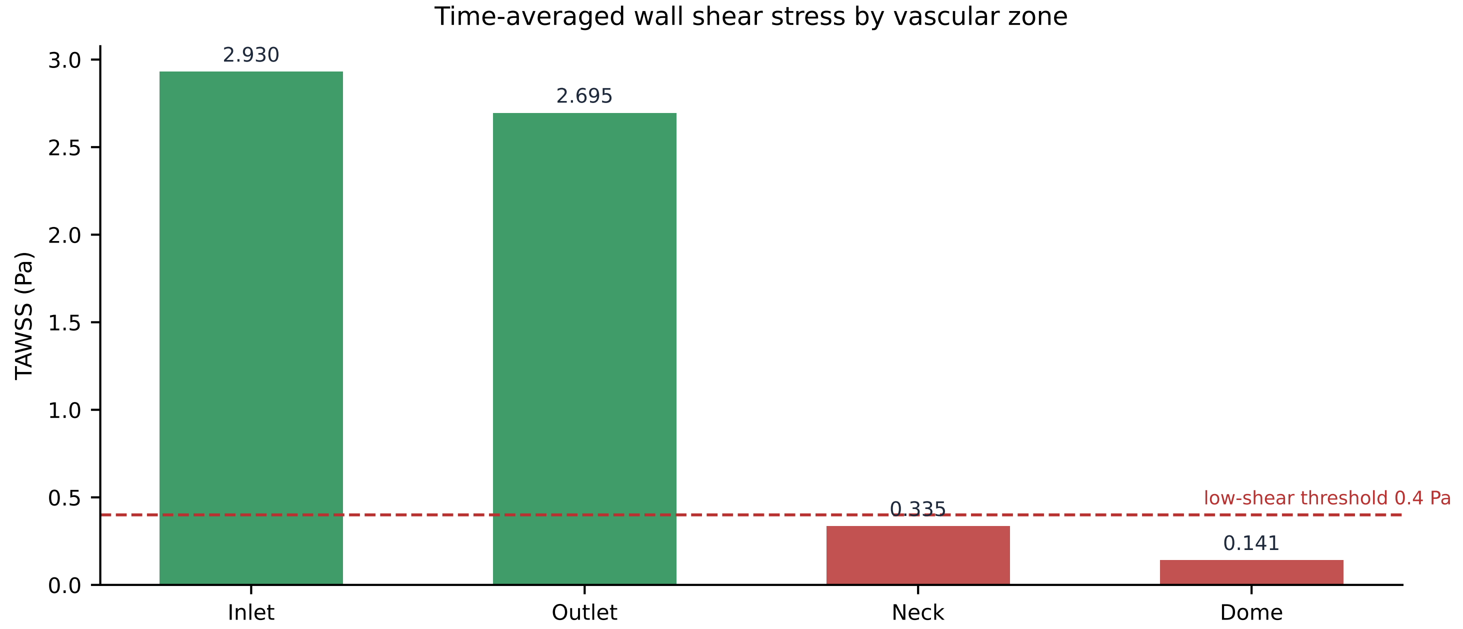
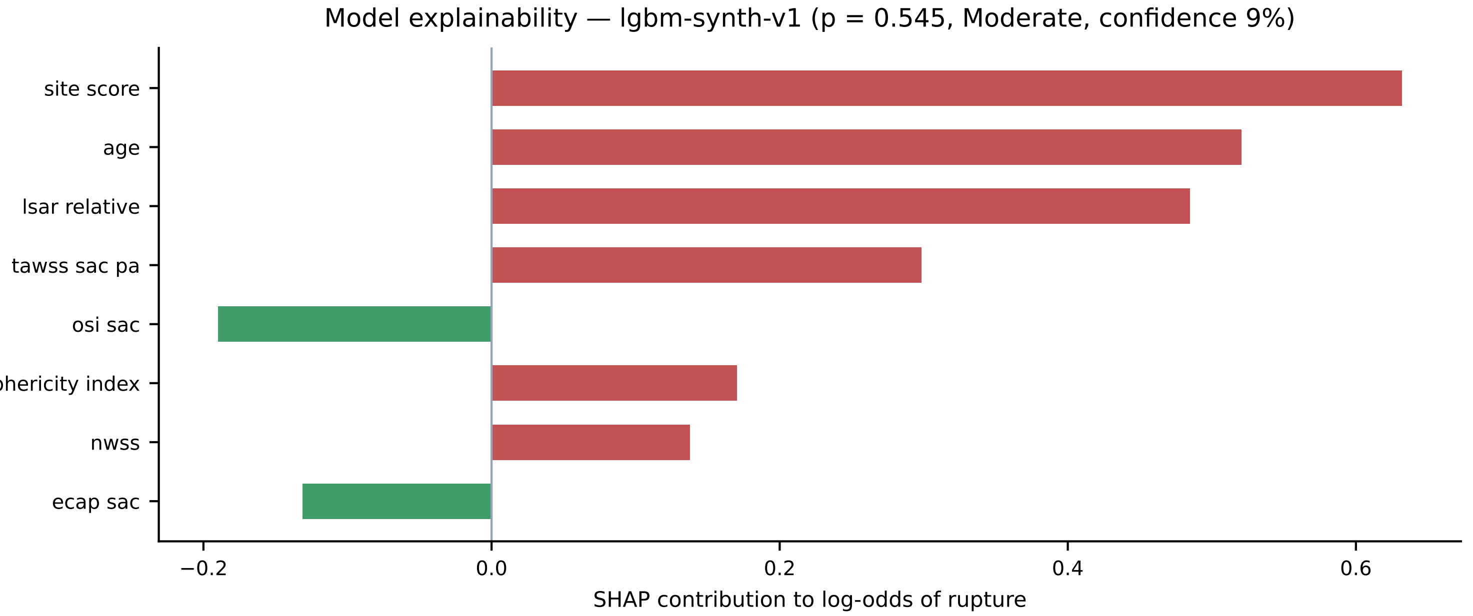
Zone Analysis and Model Explainability

Case PT-2026-0102





Assessment

Incomplete input vector: OSI and ECAP were not computed for this steady solve and entered the model as zero.

Computational fluid dynamics analysis of case PT-2026-0102. Time-averaged wall shear stress at the aneurysm dome measured 0.141 Pa against 2.930 Pa in the parent artery (normalised WSS 0.048), with an oscillatory shear index of 0.000. 92.0% of the sac surface lies below 10% of the parent-artery shear (LSAR). Relative residence time at the dome is 21.35 Pa^-1. Sac morphology measured from the reconstructed surface: maximum diameter 11.02 mm, neck 8.22 mm, aspect ratio 1.12, volume 665.6 mm^3. Composite Risk Index 59/100 (Moderate).

Model caveat: ILLUSTRATIVE ONLY. Not trained or validated on patient data. Demonstrates the feature/inference/explainability pipeline. Must not be used for clinical decision-making.